Topological Magnon Nodal-Lines in Layered Collinear Antiferromagnets

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2. Theoretical Framework
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What are Magnons?

The Quantum Picture of Spin Waves

- In magnetically ordered materials, a local excitation (a single spin flip) propagates as a collective wave of spin precessions, known as a **spin wave**.
- Just as lattice vibrations are quantized as phonons, spin waves are quantized as **magnons**.

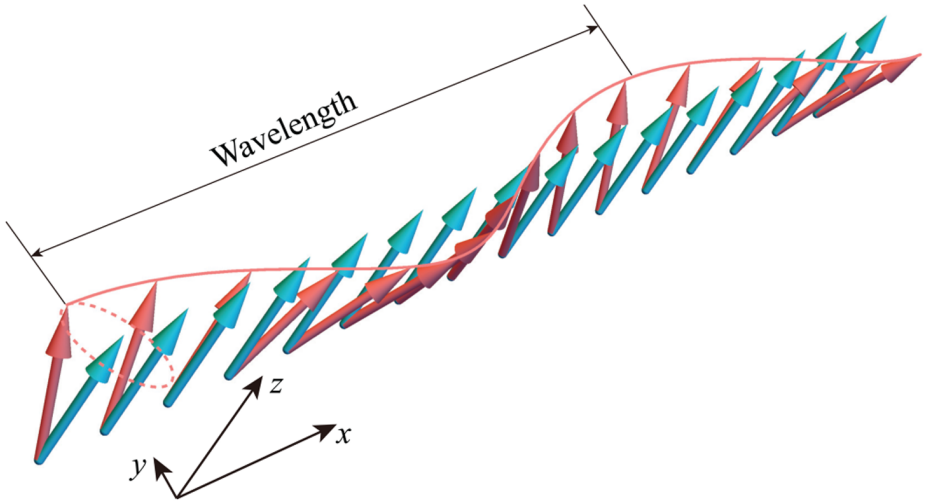

Figure: Schematic illustration for a classical spin wave.

Why are Magnons Efficient Information Carriers?

1. No Joule Heating

- Magnons carry spin angular momentum without transporting electric charge.
- This means they do not suffer from Joule heating ($P = I^2R$), a major source of energy loss in conventional electronics.

2. High Frequency Operation

- Spin dynamics in antiferromagnets naturally occur at very high frequencies, typically in the **Terahertz (THz)** range.
- This is orders of magnitude faster than the Gigahertz (GHz) clock speeds of modern electronic processors, promising a massive leap in computational speed.

The Need for Robustness: Topological Protection

The Challenge in Miniaturization

As devices become smaller, they are more susceptible to imperfections and thermal noise, which can scatter information carriers and disrupt operation.

The Topological Solution

- Topological states of matter host special "protected" states where carriers can flow without scattering from defects.
- This inherent robustness is the primary motivation for studying **Topological Magnonics**.

Topology: Electrons vs. Magnons (Part 1)

Fundamental Difference: Statistics

- **Electrons** are **Fermions**: They obey the Pauli Exclusion Principle. The ground state is a filled "Fermi sea".
- **Magnons** are **Bosons**: They do not have an exclusion principle. The ground state has zero magnons.

Topology: Electrons vs. Magnons (Part 2)

Consequences for Topology

- **Electronic systems**: Topological features are only relevant near the Fermi level.
- **Magnonic systems**: Since magnons are thermally excited, topological features across the entire band structure can be populated and contribute to physical properties.

Similarities

The mathematical framework of topology (Berry curvature, Chern numbers, etc.) applies to both.

Why This System? Layered Honeycomb AFMs

A Perfect Platform

- **Honeycomb Lattice:** Naturally hosts Dirac physics (like graphene).
- **Antiferromagnet (AFM):** Provides the benefits of THz dynamics and robustness to external fields.
- **Layered (3D) Structure:** The third momentum dimension (k_z) is crucial for realizing nodal **lines**, a richer topological object than the nodal points found in 2D systems.

Probing the Band Structure: High-Symmetry Paths

Why These Paths?

- The magnon energy $E(k)$ is a complex function in 3D momentum space.
- To visualize it, we plot the dispersion along paths connecting points of high symmetry in the Brillouin Zone (e.g., Γ, K, M, H).
- Band extrema and degeneracies (like our DNLs) are guaranteed by symmetry to occur at these special points or paths.

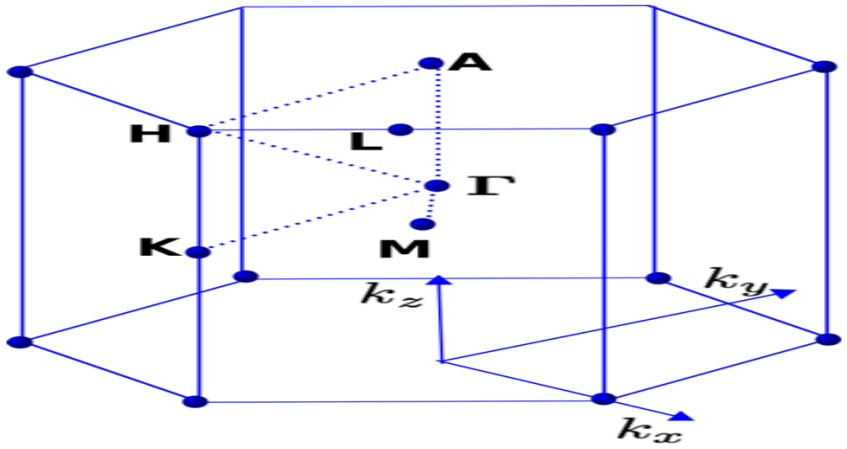

Figure: 3D bulk Brillouin zone of the hexagonal lattice.

The Challenge: Spin Commutation Relations

The Problem with Spin Operators

Spin operators $(\hat{S}^x, \hat{S}^y, \hat{S}^z)$ have non-trivial commutation relations:

$$[\hat{S}^x, \hat{S}^y] = i\hbar\hat{S}^z \tag{1}$$

This makes the many-body spin Hamiltonian problem intractable to solve directly.

The Goal

We need a method to map spin operators to simpler operators, like those for bosons, whose algebra is well understood.

The Solution: Holstein-Primakoff Transformation

From Spins to Bosons

The Holstein-Primakoff (HP) transformation maps spin operators to bosonic creation ($a^\dagger$) and annihilation (a) operators. This is valid at low temperatures, where spin deviations from the ordered ground state are small.

The Core Idea

The HP transformation correctly represents the finite spin algebra within an infinite bosonic Hilbert space.

HP Transformation for Antiferromagnets (Sublattice A)

Sublattice A (Spins Aligned Along +z)

$$S^z_{i,A} = S - a_i^\dagger a_i$$

$$S^+_{i,A} = \sqrt{2S - a_i^\dagger a_i}\ a_i \approx \sqrt{2S} a_i$$

$$S^-_{i,A} = a_i^\dagger \sqrt{2S - a_i^\dagger a_i} \approx \sqrt{2S} a_i^\dagger$$

HP Transformation for Antiferromagnets (Sublattice B)

Sublattice B (Spins Aligned Along -z)

$$S^z_{j,B} = -S + b_j^\dagger b_j$$

$$S^+_{j,B} = b_j^\dagger \sqrt{2S - b_j^\dagger b_j} \approx \sqrt{2S} b_j^\dagger$$

$$S^-_{j,B} = \sqrt{2S - b_j^\dagger b_j}\ b_j \approx \sqrt{2S} b_j$$

The Low-Temperature Approximation

The "Large-S" Limit

At low temperatures ($T \ll T_{Nel}$), the number of excited magnons is small: $\langle a_i^\dagger a_i \rangle \ll S$. This allows us to expand the square root and keep only the leading terms:

$$\sqrt{2S - a_i^\dagger a_i} \approx \sqrt{2S} \tag{2}$$

Result

The complex spin Hamiltonian is transformed into a quadratic Hamiltonian of bosons. This is the basis of linear spin-wave theory and is exactly solvable.

From Bosons to Bands: The BdG Hamiltonian

Bogoliubov-de Gennes (BdG) Formalism

After the HP transformation and a Fourier transform, the quadratic bosonic Hamiltonian takes a special matrix form known as the Bogoliubov-de Gennes (BdG) Hamiltonian:

$$\mathcal{H}(k) = \begin{pmatrix} \mathcal{A}(k) & \mathcal{B}(k) \\ \mathcal{B}^*(-k) & \mathcal{A}^*(-k) \end{pmatrix} \tag{3}$$

This structure is characteristic of systems with particle-hole symmetry, common in superconductivity and bosonic systems.

Diagonalization: Paraunitary Transformation

The Bosonic Analogue of Unitary Transformation

- For fermions (like electrons), we use a unitary transformation to diagonalize the Hamiltonian matrix.
- For bosons, due to their commutation relations, we must use a **paraunitary transformation** $\mathcal{U}_k$.
- This transformation preserves the bosonic commutation relations while finding the eigenmodes (the magnon bands).

The Result

The paraunitary diagonalization of the BdG Hamiltonian gives us the final energy dispersion relation, $E_{\pm}(k)$.

The Microscopic Spin Hamiltonian

Our model for the layered honeycomb AFM:

$$\mathcal{H} = \underbrace{J \sum_{\langle ij \rangle, l} S_{i,l} \cdot S_{j,l}}_{\text{Intralayer AFM}} + \underbrace{D \sum_{\langle\langle ij \rangle\rangle, l} D_{ij,l} \cdot S_{i,l} \times S_{j,l}}_{\text{DMI}} + \underbrace{J_c \sum_{\langle ll' \rangle, i} S_{i,l} \cdot S_{i,l'}}_{\text{Interlayer AFM}} \tag{4}$$

J: Nearest-neighbor antiferromagnetic exchange.

D: Next-nearest-neighbor Dzyaloshinskii-Moriya interaction.

J_c: Interlayer antiferromagnetic exchange.

Magnon Dispersion Relation

After applying spin-wave theory, the magnon energy bands are found:

$$E_{\pm}(k) = f^{D}(k_{||}) + \sqrt{f_0^2 - [f(k_z) \pm |f(k_{||})|]^2} \tag{5}$$

Physical Meaning of Terms

- f_0: On-site energy offset from classical exchange fields.
- $f(k_{||})$: In-plane magnon hopping amplitude.
- $f(k_z)$: Inter-plane magnon hopping amplitude.
- $f^{D}(k_{||})$: Energy shift due to the DMI.

Magnon Dispersion: The Topological DNL Phase

This plot shows the magnon dispersion relations for the original layered honeycomb antiferromagnet.

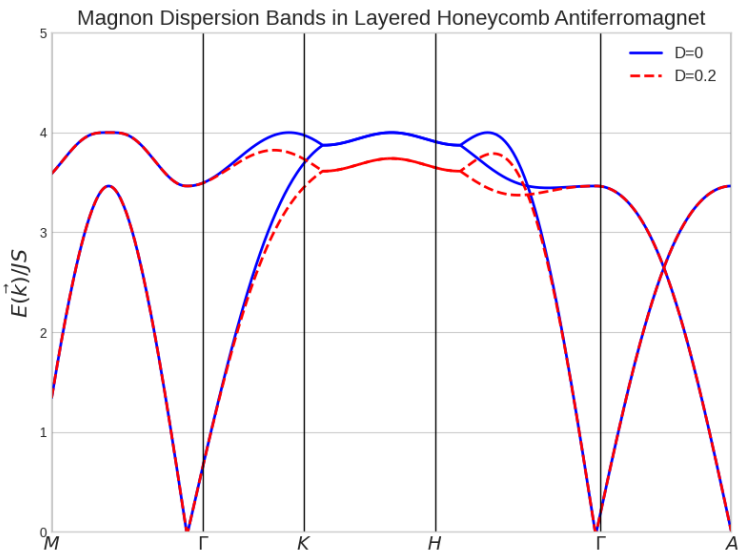

Figure: Magnon dispersion bands for (blue solid) no intralayer DMI (D=0) and (red dashed) with intralayer DMI (D=0.2). Both cases exhibit Dirac Nodal Lines along K-H

Condition for Nodal Lines

How DNLs Form

The two magnon bands become degenerate ($E_+ = E_-$) when:

$$f(k_z) \cdot |f(k_{||})| = 0$$

This condition is satisfied along continuous **lines** in the 3D Brillouin Zone, forming the Dirac Nodal Lines.

The Combined PT Symmetry

The Protecting Symmetry

The collinear AFM ground state is invariant under the combined operation of Space-Inversion (P) and Time-Reversal (T).

Why it Matters

This **PT symmetry** provides the topological protection for the DNLs, making them robust.

Symmetry Analysis: DMI and PT

The DMI term is $\mathcal{H}_{DM} = \sum D \cdot (S_i \times S_j)$.

Under Time Reversal (T)

Spins reverse ($S \to -S$), so $S_i \times S_j \to S_i \times S_j$. The DM vector D reverses. Thus, $\mathcal{H}_{DM} \to -\mathcal{H}_{DM}$. **T is preserved**.

Under Space Inversion (P)

The vectors S_i and S_j do not change, but the vector D (which depends on the bond orientation) reverses sign. Thus, $\mathcal{H}_{DM} \to -\mathcal{H}_{DM}$. **P is broken**.

Result

Since the DMI term is odd under T, it preserves the PT symmetry of the AFM ground state. Therefore, it cannot gap out the DNLs.

Probing Topology with Hall Effects

What is a Hall Effect?

A Hall effect is a transverse response to a longitudinal driving force. It is a powerful probe of a material's intrinsic quantum geometric and topological properties.

Magnon Thermal Hall Effect

The magnon thermal Hall effect is a transverse heat current generated by a longitudinal temperature gradient in a magnetic insulator. It arises from the Berry curvature of magnon bands, which can be induced by Dzyaloshinskii-Moriya interactions or external magnetic fields.

Significance

The magnon thermal Hall effect provides insights into the topological nature of magnon bands and can be used to identify topological phases in magnetic materials.

Berry Curvature and The Chern Number

Berry Curvature: $\Omega(k)$

A geometric property of the band structure that acts like a momentum-space magnetic field.

$$[\Omega_{\alpha\beta}(k)]_{nn} = -2\Im[\tau_3(\partial_{k_\alpha}\mathcal{U}_k^\dagger)\tau_3(\partial_{k_\beta}\mathcal{U}_k)]_{nn} \tag{6}$$

Chern Number: C_n

Integrating the Berry curvature over a 2D surface gives a quantized topological invariant.

$$\mathcal{C}_n = \frac{1}{2\pi}\int_{BZ} dk_{||}\Omega^n_{xy}(k) \tag{7}$$

How PT Symmetry Forces Vanishing Berry Curvature

The combined PT symmetry imposes two conflicting mathematical constraints on the Berry curvature, $\Omega(\vec{k})$

- **Time-Reversal ($\mathcal{T}$)** requires the Berry curvature to be an **odd function** of momentum

$$\Omega(\vec{k}) = -\Omega(-\vec{k})$$

- **space-Inversion ($\mathcal{P}$)** requires the Berry curvature to be an **even function** of momentum:

$$\Omega(\vec{k}) = \Omega(-\vec{k})$$

The only way for a function to be both odd and even simultaneously is for it to be identically zero everywhere.

Result

Combining the two constraints leads to the inevitable conclusion.

$$\Omega(\vec{k}) = -\Omega(-\vec{k}) = -\Omega(\vec{k}) \quad \Longrightarrow \quad \Omega(\vec{k}) = 0$$

Key Result: Vanishing Linear Response

A Consequence of PT Symmetry

In the DNL phase, the protecting PT symmetry forces the Berry curvature to be identically zero everywhere:

$$\Omega(k) = 0$$

Physical Consequence

The Magnon Thermal Hall Effect (κ_{xy}) and the Magnon Spin Nernst Effect (α^s_{xy}) are therefore strictly **absent**.

Breaking PT with an External Magnetic Field

The Zeeman term is $\mathcal{H}_H = -H \sum S^z$.

Under Time Reversal (T)

$S \to -S$, so $\mathcal{H}_H \to -\mathcal{H}_H$. **T is broken**.

Under Space Inversion (P)

The spin operator S^z is an axial vector, which is even under inversion. $\mathcal{H}_H \to \mathcal{H}_H$. **P is preserved**.

Result

Since T is broken, the combined PT symmetry is broken. The symmetry protecting the DNLs is now lifted, and a gap opens.

Topological Phase Transition: Magnon Chern Insulator

Properties of the New Phase

The gapped system is now a **3D Magnon Chern Insulator**. Its bands are characterized by a non-zero Chern number, indicating non-trivial topology ($C = \mp 1$).

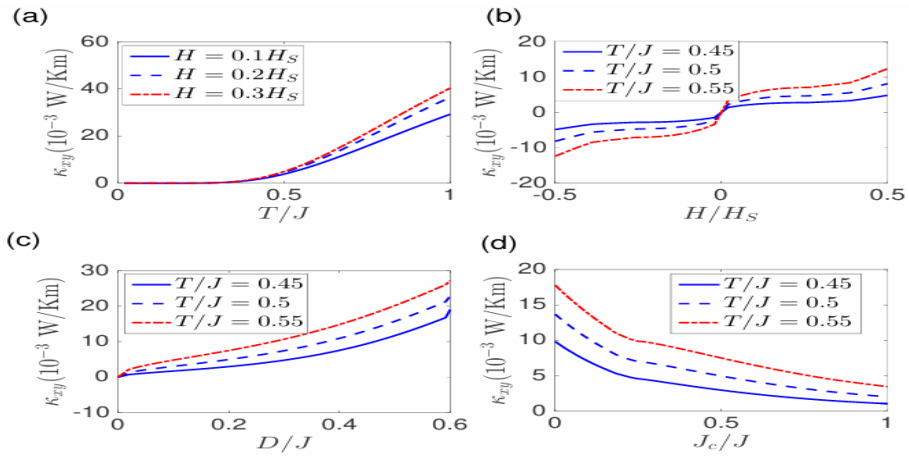

Figure: Predicted magnon thermal Hall conductivity for our system as a function of (a) T/J for different values of magnetic field, (b)H/H_S, (c) D/J, and (d)J_c/J for different values of temperature[1].

Experimental Signature: Thermal Hall Effect

Measurable Consequence

The now non-zero Berry curvature gives rise to a measurable **Magnon Thermal Hall Effect** (κ_{xy}).

$$\kappa_{xy} = -k_B T \int_{BZ} \frac{dk}{(2\pi)^3} \sum_{n=1}^{N} c_2[f_n^{BE}(k)]\Omega_{\alpha\beta}^n(k) \tag{8}$$

This serves as a key experimental signature of the field-induced topological phase.

New Proposal: An Inter-layer Spin-Canting Interaction

We introduce a new interaction term, $\mathcal{H}_{D_c}$, which represents a structurally-derived chiral interaction between next-nearest-neighbor (NNN) spins on adjacent layers.

$$\mathcal{H}_{D_c} = \sum_{\langle i,j \rangle, \langle l,l' \rangle} \vec{D}_c \cdot (\vec{S}_{i,l} \times \vec{S}_{j,l'})$$

- The vector $\vec{D}_c = D_c \hat{z}$ is fixed by the crystal geometry (stacking axis), not by spin-orbit coupling.
- This term energetically favors a non-collinear, canted (or twisted) arrangement of spins between the layers.
- **The key question:** How does this new term transform under the protecting PT symmetry?

Explicit Breaking of PT Symmetry

We analyze the transformation of $\mathcal{H}_{D_c}$ under $\mathcal{P}$ and $\mathcal{T}$ individually.

1. **Action of space-Inversion ($\mathcal{P}$):**
   - The DMI vector is a polar vector: $\mathcal{P}(\vec{D}_c) = -\vec{D}_c$.
   - The cross product of spins is an axial vector: $\mathcal{P}(\vec{S} \times \vec{S}') = +(\vec{S} \times \vec{S}')$.
   - Thus, $\mathcal{P}(\mathcal{H}_{D_c}) = (-\vec{D}_c) \cdot (+(\vec{S} \times \vec{S}')) = -\mathcal{H}_{D_c}$ (The term is odd).

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2. **Action of Time-Reversal ($\mathcal{T}$):**
   - The DMI vector is structurally derived and static: $\mathcal{T}(\vec{D}_c) = +\vec{D}_c$.
   - The spin cross product has two sign flips: $\mathcal{T}(\vec{S} \times \vec{S}') = (-\vec{S}) \times (-\vec{S}') = +(\vec{S} \times \vec{S}')$.
   - Thus, $\mathcal{T}(\mathcal{H}_{D_c}) = (+\vec{D}_c) \cdot (+(\vec{S} \times \vec{S}')) = +\mathcal{H}_{D_c}$ (The term is even).

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   - Thus, $\mathcal{T}(\mathcal{H}_{D_c}) = (+\vec{D}_c) \cdot (+(\vec{S} \times \vec{S}')) = +\mathcal{H}_{D_c}$ (The term is even).
3. **Combined Action ($\mathcal{PT}$):**

$$\mathcal{PT}(\mathcal{H}_{D_c}) = \mathcal{P}(\mathcal{T}(\mathcal{H}_{D_c})) = \mathcal{P}(+\mathcal{H}_{D_c}) = -\mathcal{H}_{D_c}$$

Since $\mathcal{PT}(\mathcal{H}_{D_c}) \neq +\mathcal{H}_{D_c}$, the new interaction term **breaks the protecting PT symmetry**.

Consequence: Loss of the DNL Phase

Because the interlayer spin-canting interaction breaks PT symmetry, it is no longer forbidden from lifting the band degeneracy. This drives a topological phase transition.

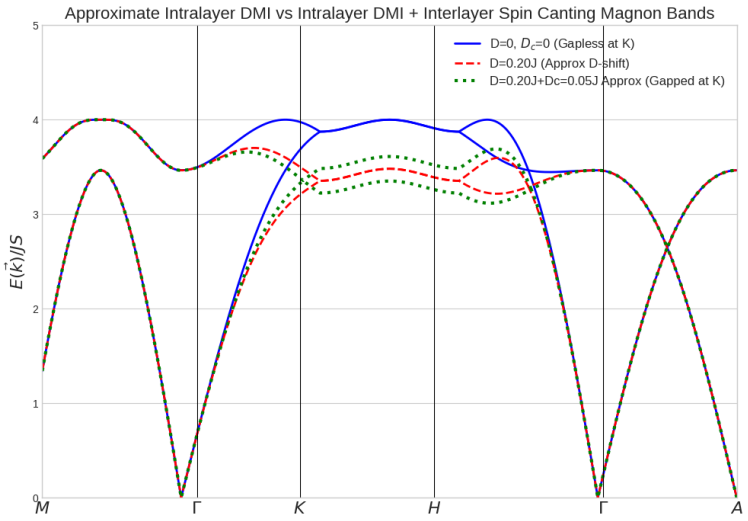

Figure: The green dotted lines (New Term) show the dispersion for the model including the interlayer interaction (D_c). A clear **gap** has opened between the bands (e.g., along the K-H path), confirming the destruction of the DNL phase.

A Special Case: The $k_z = \pi/2$ Plane

The symmetry breaking is subtle and momentum-dependent. The effect of the new term in momentum space is described by the function:

$$f^{D_c}(k) \propto \cos(k_z)$$

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- This means the symmetry-breaking interaction **vanishes entirely** within this 2D subspace of the Brillouin Zone.
- **Conclusion:** In the $k_z = \pi/2$ plane, the PT symmetry is effectively restored, and the Dirac Nodal Lines (or Dirac points in 2D) remain gapless and protected.

Effect of Interlayer Spin Canting Interaction

An interlayer DMI term, $\mathcal{H}_{D_c}$, introduces a new term in the Hamiltonian:

$$f^{D_c}(k) = 2D_c S \cos(k_z) \frac{|f(k_{||})|}{JS} \tag{9}$$

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A Subtle Result

Even though this term breaks PT symmetry globally, its effect vanishes at the special $k_z = \pi/2$ plane because $\cos(\pi/2) = 0$. The DNL remains protected and gapless within this specific 2D momentum-space plane.

Summary: Key Findings

- Layered honeycomb AFMs host **Dirac Nodal Line magnons** protected by **PT symmetry**.
- This symmetry forces the linear Hall responses $(\kappa_{xy}, \alpha^s_{xy})$ to be strictly zero.
- Breaking PT symmetry with an external magnetic field induces a topological phase transition to a **Magnon Chern Insulator**.
- This gapped topological phase has a clear experimental signature: a non-zero **Magnon Thermal Hall Effect**.

Further Motivations and Future Directions

Nonlinear Responses and Quantum Geometry

Remarkably, the system can still exhibit an intrinsic **nonlinear Hall response** arising from its quantum geometric properties. This effect highlights how nontrivial band geometry can generate transverse magnonic transport without explicit symmetry breaking.

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Future Investigations

- Further investigation into nonlinear magnon transport in layered antiferromagnets.
- Probing materials which exhibit such structures to find marks of this non-linear Hall response.

Experimental Realizations

Candidate Materials

This theory is not purely abstract. We plan on working further and finding out potential real material candidates which can host these Symmetry-protected topological DNL Magnons.

Usefulness

Verifying these topological properties in real materials would be a major step towards creating practical, low-power magnonic devices for information processing and storage.

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